7. Python Program to Implement BFS (Breadth-First Search)

Aim: To implement Breadth-First Search (BFS) in Python to traverse a graph level by level.

Procedure:

1. Define a graph using an adjacency list.
2. Select the starting node.
3. Create a queue to store nodes for traversal.
4. Mark the starting node as visited.
5. Remove a node from the queue and visit its unvisited neighbours.
6. Continue until the queue is empty and display the traversal order.

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

def bfs(start):
    visited = set()
    queue = deque([start])
    visited.add(start)

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

bfs('A')
```

Output:

```
Python 3.14.7 (tags/v3.14.7:823f032,
AMD64)] on win32
Enter "help" below or click "Help" ab
>>
============== RESTART: C:/Users/ace
A B C D E F
>>
```

Result: Thus, Breadth-First Search was implemented successfully using Python, and the graph was traversed level by level.